

The Race between Man and Machine

Tasks, wages, and what a formal proof can establish

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<https://github.com/johnbarraza/ai-06-restrepo>

Road map for 20 minutes

- ① The aggregate task economy and equilibrium threshold
- ② Automation versus new tasks: opposite reallocations
- ③ The wage condition, its analytical and numerical check
- ④ Lean translation: checked theorem, proof step, and limits
- ⑤ Hand verification and verdict

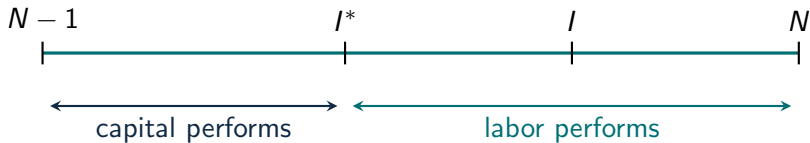
Question and unit of analysis

Question

Can automation raise output yet lower the wage? Can new tasks reverse the fall in labor's share?

Unlike the earlier weekly papers, the unit here is the **aggregate economy**: a continuum of production tasks and competitive factor markets.

The task space



Two frontiers

I is technological feasibility; I^* is actual capital use. They need not coincide.

From tasks to aggregate output

$$Y = \tilde{B} \left(\int_{N-1}^N y(i)^{(\sigma-1)/\sigma} di \right)^{\sigma/(\sigma-1)}.$$

- ▶ The task interval always has measure one.
- ▶ $\sigma > 0$ governs substitution across tasks; the $\sigma = 1$ case is understood by its limit.
- ▶ Automation changes *who performs* tasks; new task creation shifts the interval to higher i .

The resulting aggregate production function has technology-dependent capital and labor weights (Proposition 1).

The agents' problems

Competitive task producer For $i \leq I$,
compare factor costs

$$\min \left\{ R, \frac{W}{\gamma(i)} \right\}.$$

For $i > I$, only labor is feasible. Intermediate inputs also enter full unit cost.

Representative household Choose (C, L)
with $C = WL + RK$ to maximize the paper's
 $u(C, L)$. At an interior choice:

$$\nu'(L) = \frac{W}{C}.$$

Comparative advantage implies a threshold

$$\frac{W}{R} = \gamma(\tilde{I}), \quad I^* = \min\{I, \tilde{I}\}.$$

- ▶ $\gamma(i)$ strictly increases: labor has an advantage in more complex tasks.
- ▶ Below I^* : capital is cheaper; above I^* : labor is used.
- ▶ If $I < \tilde{I}$, technology constrains automation.
- ▶ If $\tilde{I} < I$, technology is slack; more feasible automation changes nothing locally.

Conditions behind the static results

- ① $\gamma(i)$ strictly increasing (Assumption 1).
- ② $\eta \rightarrow 0$ or $\zeta = 1$ (Assumption 2): homothetic factor demand.
- ③ $K < \bar{K}$ (Assumption 3): new tasks are used.

$\sigma > 0$, $\hat{\sigma} > 0$, $\varepsilon_L > 0$, and the household's convexity and regularity conditions also matter. Proposition 1 supplies a unique static equilibrium under these conditions.

Automation and reinstatement

Automation: $I \uparrow$

Capital takes over some existing labor tasks. In the constrained regime, W/R , labor share, and employment fall.

New tasks: $N \uparrow$

New complex tasks require labor and replace low-index tasks. W/R , labor share, employment, and the wage rise.

These are Proposition 2–3 static comparative statics, not unconditional predictions for a dynamic economy.

Where the wage equation comes from

$$(B9) \quad s_L d \ln W + (1 - s_L) d \ln R = d \ln Y|_{K,L},$$

$$(B10), dN = 0 : \quad d \ln W - d \ln R = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dl.$$

Let $x = d \ln W$, $z = d \ln R$, and $p = d \ln Y|_{K,L}$. Since $z = x - (x - z)$, (B9) becomes

$$x = p + (1 - s_L)(x - z).$$

Substituting (B10) gives the Proposition 3 wage decomposition. This two-equation algebra is the natural Lean target.

The wage trap: two forces

$$\underbrace{\frac{d \ln W}{dl}}_{\text{wage response}} = \underbrace{P_I}_{\text{productivity gain}} - \underbrace{\frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{\text{displacement}}.$$

Exact local condition, $I^* = I < \tilde{I}$

The wage rises iff

$$P_I > \frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

Equality gives no local change; a reversed inequality gives a fall.

A different sign is fixed

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_l}{\hat{\sigma} + \varepsilon_L} < 0.$$

The labor share and employment move with $\omega = W/(RK)$ and therefore fall in the constrained case.

Interpretation

A falling labor share does not prove that the *level* of the real wage falls. Productivity and displacement compete in the wage equation.

Analytical and computational check

Illustrative case	P_I	displacement	$d \ln W / dI$
Large productivity gain	.400	.133	+.267
Small productivity gain	.050	.133	-.083
Equality	.133	.133	0

Same $s_L = .6$, $\Lambda_I = .5$, $\hat{\sigma} = 1$, $\varepsilon_L = .5$ in every row. Reproduce with python `analysis/wage_condition.py`. These inputs illustrate the sign identity; they are not solved equilibria.

Where I did not trust the AI

Claim to test

“Automation necessarily reduces wages and labor’s share.”

- ▶ Labor share falls only when the automation frontier binds; at a slack frontier, a small increase in I has no effect.
- ▶ The wage can rise when the productivity gain dominates displacement.
- ▶ At $I^* = I = \tilde{I}$, use one-sided derivatives.

Hand check: the decisive inequality

Real photograph of the student's handwritten derivation pending. See `hand/README.md`.

$d \ln W / dI = P_I - D_I$, so a wage increase requires $P_I > D_I$.

Lean run: source to statement

Pending run

The own-run formalization is in progress. This slide will show an actual source claim, exact Lean statement, and proof fragment after the check; no placeholder theorem is presented as verified.

What the checks can and cannot conclude

- ▶ The computational script verifies arithmetic on the stated wage identity.
- ▶ Lean's compiler can verify a formal theorem only under its encoded assumptions and domain.
- ▶ Matching the NBER claim requires a separate source-to-Lean comparison.
- ▶ The full continuum equilibrium and all its comparative statics should be counted only if formally represented and proved.

Conclusion

Economic result

Automation reallocates tasks away from labor. With a binding frontier, labor share falls, while wages depend on the balance between productivity and displacement. New tasks reinstate labor.

Methodological result

The threshold, regime conditions, and sign inequality must all be explicit in any Lean translation or oral explanation.

Sources

Acemoglu, D., and P. Restrepo (2018). “The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment.” *AER* 108(6), 1488–1542.

Formalization source: NBER Working Paper 22252, revised June 2017, 87 pages. In particular Section 2, Propositions 1–3 and Appendix B.

Course requirement: <https://github.com/alexanderquispe/AI-Econ-Modeling/issues/5>